

OCR Further Mathematics
Statistics - Non-parametric Tests
Mark Scheme
AS and A Level
Further Mathematics A - H235, H245



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The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
Students of other boards may also find this useful.



Question			Answer/Indicative content	Marks	Part marks and guidance	
1	a		H_0: Two samples are from identical populations H_1: Two samples are from populations with different median ratings. $R_m = 1 + 2 + 3 + 4 + 5 + 9 + 10 + 11 (= 45)$ $W = 45$ $8(8 + 8 + 1) - R_m = 91$ $W_{crit} = 49$ Reject H_0. Significant evidence that there is a difference in median ratings / opinions have changed	B2 (AO1.1) (AO1.1) M1 (AO1.1) A1 (AO1.1) B1 (AO2.1) B1 (AO1.1) M1ft (AO1.1) A1ft (AO2.2b) [8]	If no reference to "populations", maximum B1 Allow H_0: "identical population medians", H_1: "not identical populations" or "not identical pop medians" "Pupils' opinions have not changed", etc: B2 If omitted, can still get all other marks FT on TS (< 68) or CV FT on TS only. Allow "increased" SC: Sign or paired-sample test, max B2 (hypotheses)	
	b		Eliminate the difference between individual pupils' opinions	B1 (AO3.5b) [1]	"Minimises the difference in tastes" B1 (BOD) Scores arbitrary: B1 (etc). Not "more powerful test".	
	c		A paired-sample signed-rank test would have been used	B1 (AO3.5c) [1]	Must mention "paired sample" oe - not just "Wilcoxon signed rank"	
	d		$0.025 \times 12870 = 322$	M1 (AO3.1a) A1 (AO3.2a) [2]	$0.05 \times 12870 = 643.5$ M1 321 or 322 or 643 (from 1-tail), must be integer	
			Total	12		
2	a		$H_0: m_A = m_B$, $H_1: m_A < m_B$ where m_A and m_B are the median journey times for A and B $W \sim N(180, 510)$	B1 (AO1.1) B1 (AO1.1) M1	OR: Median journey times equal, oe. Allow if ms used but not defined Both, can Allow "mean" or "average" only if "population" stated Allow $\sqrt{510}$	



Question			Answer/Indicative content	Marks	Part marks and guidance		
			Consider correct tail, either 219 or 141 $(R_m = 219, m(m + n + 1) - R_m = 141)$ $p = \Phi\left(\frac{141.5-180}{\sqrt{510}}\right) = 0.0441 \dots$ BC $0.0441 < 0.1$ 				



Question			Answer/Indicative content	Marks	Part marks and guidance		
	b		Must be a random sample (of all journeys) Or distributions must be same shape (necessary assumption for Wilcoxon rank-sum test!)	B1 (AO3.5b) [1]	Or equivalent. Allow “(journeys) independent”	Not “representative”.	
			Total	9			
3	a		Test is for rankings / rankings arbitrary / not bivariate normal etc	B1 (AO2.4) [1]	OE		
	b		$H_0: \rho_s = 0$, $H_1: \rho_s > 0$, where ρ_s is the population rank correlation coefficient Rank 5 4 3 6 1 2 s 5 1 2 6 4 3 $\Sigma d^2 = 20$ $r_s = 1 - \frac{6 \times 20}{6 \times 35}$ = 3/7 or 0.42857... < 0.9429 Do not reject H_0 Insufficient evidence of association between ranking given by the two categories	B1 (AO1.1) B1 (AO1.1) M1 (AO1.1) A1 (AO1.1) B1 (AO1.1) M1ft (AO1.1) A1ft (AO2.2b) [7]	Allow ρ_s not defined; allow ρ. Allow: H_0: no association between rankings. H_1: positive association (but <i>not</i> H_1: association) Exact or awrt 0.429 FT on their Σd^2 only		
	c		Not dependent on any distributional assumptions	B1 (AO1.2) [1]	Oe (cf. Specification, 5.08f)		
			Total	9			



Question	Answer/Indicative content	Marks	Part marks and guidance		
4	$H_0: m_Q = m_R, H_1: m_Q \neq m_R$, where m_Q and m_R are the medians of the rankings given to Q and R Sum of ranks = $\frac{1}{2} \times 54 \times 55 = 1485$ $R_m = 1485 - 726 = 759$ [or 561] $R_m \sim N(660, \dots 3300)$ $P(R_m \geq 759) = 0.0432$ (3 s.f.) [or $z = 1.715$] Alternatively: CV $660 + 1.96\sqrt{3300} [= 772.6]$ $758.5 < 772.6$ $p > 0.025, 2p > 0.05, z < 1.96$, or 1.96 used in CV Do not reject H_0. Insufficient evidence of a difference between the ranks. SC 726 or 594 used: can get B1; M0A0; M1A1; M1A0 [0.1235, 0.1253 or 0.1272]; B1M1A1, total 7/10	B1(AO1.1) M1(AO1.1) A1(AO1.1) M1(AO3.1b) A1(AO3.3) M1(AO3.4) A1(AO1.1) M1 A1 B1(AO1.1) M1ft(AO1.1) A1ft(AO2.2b) [10]	Allow m undefined. If verbal, must mention medians, m or distribution. Allow $m_d = 0$ as opposed to $m_Q = m_R$ Find sum of ranks Correct value of R_m seen normal, mean their $\frac{1}{2} \times 24 \times 55$ Both parameters correct Standardise, their R_m Correct test statistic (0.0432) or 0.0424 or 0.0416 (no / wrong cc): M1A0 Not 759 - or 726 - ...; not wrong tail for comparison, but allow $\pm$ Needs correct cc Explicit correct comparison Correct first conclusion,	Not anything that might be μ unless symbol clearly defined as median. Not "there is no difference in the ranks ..." Allow even if 726 used later Allow SD / Var muddle (Same for $P(R_m < 561)$) Allow $z \in [1.71, 1.715]$, allow $z = 1.72$ only if cc demonstrate correct Or $561.5 > 547.4$ Wrong z -value: M1A1ft B0 Needs like-with-like (e.g. p must be < 0.5) ft on wrong	



Question			Answer/Indicative content	Marks	Part marks and guidance	
			For 726 and N(825, 3300) giving $p = 0.0432$: B1; M0A0; M0A0; M1A1; B1M1A1, total 6/10		needs correct method and like-with-like Co contextualise d, not too definite ts, or 1-tail/2-tail confusions, e.g. p compared with 0.05 or not explicit, or $z \geq 1.645$ <u>Examiner's Comments</u> Good statements of the hypotheses involved the medians of the rankings, or more generally their distributions. However, many stated their hypotheses as 'the ranks of the candidates are the same', which is incorrect; it suggests that all the candidates had exactly the same rankings, which is obviously either true or false.  Misconception Hypotheses always refer to an underlying distribution, never to the actual data values. Hypothesis tests use data values to make inferences about the underlying distributions. About half the candidates realised that they needed not the sum of the rankings of candidates taught by Professor Q, but the sum of the rankings of the <u>smaller</u> group of the data, in other words those taught by Professor R. It was therefore necessary to find the sum of all 54 rankings	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
					(which is 1485) and subtract 726 from this. However, when using a normal approximation it is <u>not</u> necessary to use the smaller of R_m and $m(m + n + 1) - R_m$, as the symmetry of the normal distribution will give the same answer for either. Some, having correctly found 759, unfortunately went on to use 726 as it is less than 759. The choice of R_m and $m(m + n + 1) - R_m$ is relevant only to the rank sum of the smaller sample and even then it is not needed when a normal approximation is used. The rule ('use the smaller of the two') is there to save space in the tables and to cope with cases when the data is ranked top-to-bottom as opposed to bottom-to-top.	
			Total	10		

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Question			Answer/Indicative content	Marks	Part marks and guidance		
5		a	$H_0: \mu = 700$ $H_1: \mu < 700$ where μ is the mean reaction time for all international athletes $\bar{x} = 607$ $z = -1.822$ or $p = 0.0342$ or $CV = 616.05...$ $z < -1.645$ or $p < 0.05$ or $607 < CV$ Reject H_0 Significant evidence that mean reaction times of international athletes are shorter	B2 (AO 1.1) (AO 1.1) M1 (AO 3.3) A1 (AO 3.4) A1 (AO 1.1) M1ft (AO 1.1) A1ft (AO 2.2b) [7]	One error, e.g. no or wrong letter, $\neq$, etc : B1 Find sample mean Correct z , p or CV Correct comparison Correct first conclusion Context, not too definite (e.g. <i>not</i> "international athletes' reaction times are shorter")	Ignore failure to define μ here Needs correct method, like-with-like; ft on their z , p or CV ft on their z , p or CV	



Question			Answer/Indicative content	Marks	Part marks and guidance		
		b	(i) Uses more information (e.g. magnitudes of differences) (ii) $H_0: m = 700$, $H_1: m < 700$ where m is the median reaction time for all international athletes Differences 2 -69 -16 14 -12 -22 s 0 5 0 Ranks +1 -3 -5 +2 -4 -6 $W_- = 18$ $W_+ = 3$ so $T = 3$ $n = 6$, $CV = 2$ Do not reject H_0. Insufficient evidence that median reaction times of international athletes are shorter	B1 (AO 2.4) [1] B1 (AO 2.5) M1 (AO 1.1) A1 (AO 3.3) A1 (AO 3.4) A1 (AO 1.1) A1ft (AO 2.2b) [6]	Same as in (i) but different letter or "median" stated Find differences from 700, rank and consider signs For W_- or W_+ For both, and T correct Correct CV In context, not too definite	FT on their T	
	Copyright © 2024 Exam Papers Practice	c	They use different assumptions	B1 (AO 2.3) [1]		Not "one is more accurate"	
			Total	15			



Question			Answer/Indicative content	Marks	Part marks and guidance		
6		a	$H_0: m_s = m_{bm}$, $H_1: m_s \neq m_{bm}$, where m_s and m_{bm} are the median population scores for the task in silence and with background music respectively. Ranks 4, 5, 7, 8, 9; 1, 2, 3, 6, 10 $R_m = 22$ (or 33) $m(m + n + 1) - R_m = 33$ (or 22) $W = 22$ and $CV = 19$ $W > 19$ so do not reject H_0 Insufficient evidence that background music affects scores	B1 (AO 2.5) M1 (AO 1.1a) M1 (AO 1.1a) A1 (AO 1.1) B1 (AO 1.1) M1ft (AO 2.2b) A1ft (AO 3.5a) [7]	Rank Sum one set of ranks Find $m(m + n + 1) - R_m$ Both needed FT on their W only FT on their W only	Allow 7, 6, 4, 3, 2; 10, 9, 8, 5, 1 If omitted can get 7/8	
		b	Eliminates differences between students	B1 (AO 2.3) [1]			
			Total	8			



Question			Answer/Indicative content	Marks	Part marks and guidance		
7		a	Distribution of times needs to be symmetric	B1 (AO 2.4) [1]			
		b	Uses more information	B1 (AO 3.5b) [1]	e.g. "uses magnitudes of differences"		
		c	$n = 32, \mu = 264$ $\sigma^2 = 2860$ $264 - 1.645\sqrt{2860} = 0.5$ $= 175.52$ $W_- \leq 175$ where W_- is the sum of the ranks corresponding to the negative differences	B1 (AO 3.3) A1 (AO 1.1a) M1 (AO 2.1) A1 (AO 1.1) A1 (AO 2.2a) A1 (AO 3.4) B1 (AO 2.4) [7]	$\mu - z\sigma$, allow σ^2 , ignore cc Correct, with correct or no cc Allow 176.02 only if later rounded down to 175 Needs any letter, 175 and $\leq$ Allow W_- and "smaller of sum of positive rankings and sum of negative rankings"	Allow e.g. $W_- < 176$ Allow any letter if properly explained	
			Total	9			



Question			Answer/Indicative content	Marks	Part marks and guidance		
8		i	G: 8 3 11 6 9 10 B: 7 5 4 2 1 $R_m = 19$ $W = 19$ {5(6+5+1) - 19 = 41} H_0: two samples come from identical pops H_1: samples do not come from identical pops "19" > 17, do not reject H_0. insufficient evidence, at 1% level, that p.s. girls are taller than p.s. boys.	B1 B1,B1 B1 M1 A1 [6]	Allow 1 set. follow through incorrect ranks for these Marks. or $m_G = m_B$ and $m_G > m_B$ two. Contextualised, not over-assertive.	If in words, must say popn.	

Examiner's Comments
Most gained full marks. This indicates that candidates gave the conclusion in a contextualised form which was not over-assertive.



Question			Answer/Indicative content	Marks	Part marks and guidance		
		ii	$N(351, 1170)$ $\frac{W + 0.5 - 351}{\sqrt{1170}} < -2.326$ $W < 270.9$ $W = 270$	B1, B1 M1 B1 A1 A1 [6]	allow M1 if wrong or no cc. Must be -ve z. 2.326 Allow 271 for this mark, but not from 271.4 allow $\leq$	Allow =.	
			Total	12	Examiner's Comments Most gained full marks, but many considered the wrong tail. A few did not use the continuity correction.		



Question			Answer/Indicative content	Marks	Part marks and guidance		
9			$F(9) = 0.0214$, $F(10) = 0.0494$ $F(20) = 0.9786$, $F(19) = 0.9506$ $k \leq 9$ $k \geq 21$	M1 M1 A1 A1 [4]	use of tables, even if 1 tail. look for 2.5% at both ends	Allow even if incorrect eg look for 0.0025. Can be implied by attempt at $k \geq 21$. May be by symmetry. $N(15, 7.5)$ M1M1A1A0 max.	
			Total	4	Examiner's Comments Many candidates gained full marks. Those who did not usually lost marks for considering only one tail. Others looked for 0.0025 rather than the correct 0.025. Some considered $N(15, 7.5)$.		

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Question			Answer/Indicative content	Marks	Part marks and guidance		
10			d: 33 -2 -3 25 46 -5 8 10 -1 14 mean = 12.5, sd = 17.213 $H_0: \mu_D = 0, H_1: \mu_D > 0$ $\frac{12.5}{17.213}$ $\frac{\sqrt{10}}{= 2.296}$ CV = 1.833 2.296 > 1.833, reject H_0, There is sufficient evidence that process 2 is an improvement. Differences are normally distributed.	B1 B1B1 B1 M1 A1 B1 M1 A1 B1 [10]	can be implied by correct mean, sd. Allow 134.1 - 121.6; $s^2 = 296.3$ or 5333/18 Allow $\mu_1 = \mu_2$ etc. Allow in words unless clearly referring to sample. Allow 2.30 TS > CV ft. cwo. Contextualised, not over-assertive. Examiner's Comments Most candidates scored at least 9 out of 10, some losing the final mark for the assumption. A few did not use a paired test. .	Not paired. B0B1B0B1 M0A0B1(1.734 or 1.833)M1A0B0 4/10max	
			Total	10			

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Question			Answer/Indicative content	Marks	Part marks and guidance																							
11		a	<table border="1"><tr><td>7</td><td>2</td><td>1</td><td>1</td><td>1</td><td>1</td><td>4</td></tr><tr><td>4</td><td>2</td><td>1</td><td>7</td><td>6</td><td>5</td><td>3</td></tr><tr><td>+</td><td>-</td><td>-</td><td>+</td><td>-</td><td>+</td><td>+</td></tr></table> H_0: population median difference = 0 H_1: population median difference $\neq 0$ $P = 4+7+5+3=19$ $Q = 1+2+6 = 9$ $T = 8$ $T_{crit} = 3; 8 > 3$ Do not reject H_0. Insufficient evidence of a difference in test scores	7	2	1	1	1	1	4	4	2	1	7	6	5	3	+	-	-	+	-	+	+	M1(AO1.1) B1(AO2.5) A1(AO3.3) A1(AO3.4) B1(AO1.1) A1FT(AO2.2b) [6]	Calculate differences, rank them and attach signs Hypotheses correctly stated P or Q correct Both P and Q seen, T correct Comparison with correct T_{crit} Correct conclusion, in context, acknowledge uncertainty FT their T but not their T_{crit}	Follow through with correct signs and ranks from incorrect differences SC3: Two-sample, max 3 / 6	
7	2	1	1	1	1	4																						
4	2	1	7	6	5	3																						
+	-	-	+	-	+	+																						
		b	Uses magnitude of differences oe	B1(AO3.5b) [1]																								
			Total	7																								

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Question			Answer/Indicative content	Marks	Part marks and guidance	
12			$H_0: m \leq (\text{or } =) 30$, $H_1: m > 30$ Diffs 3,9, - 1,5,10,2, - 4,7 Ranks 3,7,1,5,8,2,4,6 Signed ranks 5(-), 31(+) TS = 5 5 in CR, reject H_0 There is sufficient evidence that the median time for relief is more than 30 mins. Distribution is symmetrical.	B1 M1A1 A1 A1ft M1ft A1 B1	If in words, 'population' needed. M1 for attempting differences AND ranks. Ft TS, not CV CV = 5 Cwo, in context, not over-assertive. Examiner's Comments Some gave incorrect hypotheses. $H_1: m < 30$ was quite common. The numerical part of the question was answered correctly by all but two or three candidates. Those who lost the first mark for the hypotheses usually lost the mark for the conclusion as well. Most candidates gained the mark for the assumption.	
			Total	8		

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Exam Papers Practice Copyright © 2024 Exam Papers Practice	Most scored at least eight or nine. The lost mark was usually due to there being no continuity correction.
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Question			Answer/Indicative content	Marks	Part marks and guidance																															
14		i	$H_0: m_1 = m_2$ $H_1: m_2 > m_1$	B1	Allow equiv hyps using differences. If in words, needs 'population'	NOT: marks NOT papers NOT: mean																														
		i	<table border="1"><tr><td>–</td><td>+</td><td>+</td><td>+</td><td>+</td><td>–</td><td>+</td><td>+</td><td>+</td><td>–</td></tr><tr><td>9</td><td>5</td><td>12</td><td>3</td><td>8</td><td>13</td><td>7</td><td>10</td><td>15</td><td>2</td></tr><tr><td>6</td><td>3</td><td>8</td><td>2</td><td>5</td><td>9</td><td>4</td><td>7</td><td>10</td><td>1</td></tr></table>	–	+		+	+	+	–	+	+	+	–	9	5	12	3	8	13	7	10	15	2	6	3	8	2	5	9	4	7	10	1	M1,A1	1 st A1 is for correct differences.
	–	+	+	+	+		–	+	+	+	–																									
	9	5	12	3	8		13	7	10	15	2																									
	6	3	8	2	5		9	4	7	10	1																									
		i	$T^+ = 39$; $T^- = 16$; $T = 16$	A1	2 nd A1 is for correct T from correct ranks.																															
	i	CV = 10	B1																																	
	i	TS > CV, do not reject H_0	M1	ft TS, CV																																
		i	Insufficient evidence that the calculator paper was easier. oe	A1	ft TS Contextualised, not over-assertive.	NOT: difference, unless clearly 2–tail																														
Examiner's Comments																																				
Almost all candidates scored at least 6 out of 7. Those losing a mark lost it for inadequate hypotheses or conclusions. It is pleasing that most candidates now conclude by saying 'TS > CV, do not reject H_0 so there is insufficient evidence that the second paper was easier.'																																				
	Copyright © 2024 Exam Papers Practice	ii	Differences symmetrical	B1	Examiner's Comments																															
					Most candidates scored this mark, but many did not mention 'differences' in their answer and so failed to score.																															
			Total	8																																



Question			Answer/Indicative content	Marks	Part marks and guidance	
15		i	$1 + 2 + \dots + 11$	M1	M0 if followed by incorrect work.	
		i	$= 66$	A1	Examiner's Comments Almost all candidates answered this part correctly.	
		ii	(N) (132,264)	B1		
		ii	$\frac{(W + 0.5 - "132")}{\sqrt{"264"}}$	M1	Allow wrong, or no, cc.	Allow reversed if consistent OR $132(-0.5) \pm z \times \frac{\sqrt{264}}{M1}$ $z = 2.576$ or 2.58 B1 $(89.6, [173.4])$ A1
		ii	$< -$	M1*	May be earned later.	$<$ lower limit M1
		ii	2.576	B1	Allow 2.58	≤ 89 A1
		ii	Solve inequality	*M1	Or equation if final answer uses $<$ or $\leq$	Allow if lower limit only considered.
		ii	< 89.6 ($66 \leq W \leq 89$)	A1	Integer needed.	
					Examiner's Comments The most common mark was 5 out of 6. The usual errors were to omit the continuity correction and to fail to give the final answer as an integer.	
Total				8		



Question			Answer/Indicative content	Marks	Part marks and guidance	
16			$H_0: m_{II-I} = 0, H_1: m_{II-I} < 0$	B1	Allow $m_1 > m_2, m_d > 0$, etc. or in words, but needs to be in terms of parameters or population	
			II-I: 1, -14, -9, 2, -7, -8, 3, -4, -10, 6, 5, 13	M1		
			Rank: 1, -12, -9, 2, -7, -8, 3, -4, -10, 6, 5, 11	A1		
			$P = 1 + 2 + 3 + 6 + 5 = 17$	M1A1		
			$Q = 61$ so $T = 17$			
			5% CR: $T \leq 17$			
			T is inside CR so reject H_0	M1		
			There is sufficient evidence at the 5% SL that drug II is associated with fewer sneezes	A1	ft TS only. Contextualised, not over-assertive.	
					Examiner's Comments	
					Just over half the candidates gained full marks. Of those who did not, most lost a mark for hypotheses which did not refer to the population or parameter.	
Total				7		



Question			Answer/Indicative content	Marks	Part marks and guidance	
17		i	Distribution of heights may not be normal / is unknown	B1	Allow "No assumption required", but nothing else Not "groups independent" unless something else as well Examiner's Comments Only one-third of the candidates gained this mark. Most candidates wrote 'groups independent', or equivalent.	
		ii	$H_0: m_A = m_B, H_1: m_A \neq m_B$	B1	Medians. Allow words in context. Not μ unless "median" stated	
		ii	Ranks: A: 4, 8, 10, 11, 14, 15, 16, 18, 20, 21, 22 B: 1, 2, 3, 5, 6, 7, 9, 12, 13, 17, 19	B1		
		ii	$m = n = 11, R_m = 159$ or 94	B1		
		ii	Use normal approximation with mean 126.5 [= 253/2]	M1	allow $\frac{1}{2} \times 11 \times (11+11+1)$ allow $\frac{1}{12} \times 11 \times 11 \times (11+11+1)$	
		ii	Variance 231.92 [= 2783/12]	B1		
		ii	$(\alpha) P(\leq 94) = \Phi((94.5 - 126.5) / \sqrt{(231.92)})$	M1	Standardising. Allow no / incorrect cc.	
		ii	or $P(\geq 159) = 0.0178$	A1	Value	
		ii	< 0.025 and reject H_0	M1	ft TS	
		ii	$(\beta) z = (94.5 - 126.5) / \sqrt{(231.92)} = -2.101$	M1A1	Standardising; value	
		ii	< -1.96 so reject H_0	M1	ft TS	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	There is evidence that salinity affects growth	A1	Or equivalent in context. ft TS. Examiner's Comments Most candidates scored at least seven out of nine. Marks were lost for inadequate hypotheses, use of $\frac{1}{2}$ instead of $\frac{1}{12}$ in the formula for variance and use of 0.05 instead of 0.025.	
			Total	10		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
18		i	$H_0: p = \frac{1}{2}, H_1: p > \frac{1}{2}$	B1	For both. Allow any sensible hypotheses.	
		i	Find signs of differences	M1	+++++--- or vv	
		i	Obtain 7+, 3-	A1	or vv	
		i	Attempt $P(X \geq 7)$ or $P(X \leq 3)$	M1ft		Attempt to find CR. M1 (not ft)
		i	0.1719	A1ft	Allow 0.172 (0.0547 from 8+)	$X \geq 9$ or $X \leq 1$ A1 (not ft)
		i	"0.1719" > 0.05, so do not reject H_0	M1	Ft candidate's p .	"7" (or "3") not in CR, so d.n.r. H_0 .ft
		i	Insuff. evidence that type P is better.	A1	In context, not over-assertive. Cwo.	NOT "suff evidence that there is no difference between the bows."
					Examiner's Comments About two-thirds of the candidates gained full marks. Those who did not usually lost marks for hypotheses which referred to samples or an over-assertive conclusion.	
		ii	Magnitude of differences taken into account.	B1	Uses more information. More powerful.	
					Examiner's Comments Almost all the candidates gave an acceptable response.	
			Total	8		



Question			Answer/Indicative content	Marks	Part marks and guidance	
19			$H_0: m_A = m_B, H_1: m_B < m_A$ Attempt ranks 15, 1, 6, 12, 11, 13, 14; 7, 9, 3, 10, 8, 2, 5, 4 $R_m = 72$ $W = 40$ $CV = 41$ “40” < 41 reject H_0 Evidence that treatment B is more effective.	B1 M1 A1 A1 A1 B1 M1 A1	For both. Allow any sensible hypotheses. Ft TS and CV. In context, not over- assertive. Cwo.	Examiner's Comments About three-quarters of the candidates gave a fully correct response. Those who did not usually lost marks for hypotheses which referred to samples or an over-assertive conclusion. Numerical errors were very rare.
			Total	8		